



Introduction to AWS RDS

Introduction

Amazon RDS is a fully managed relational database service provided by AWS that simplifies database setup, operation, scaling, and maintenance. It Reduce administrative overhead and Improve scalability.

Managed Tasks by AWS

- OS patching
- Database updates
- Automated backups
- Monitoring
- Failover management

User Responsibilities

- Database schema
- Query optimization
- Access control

Supported Database Engines

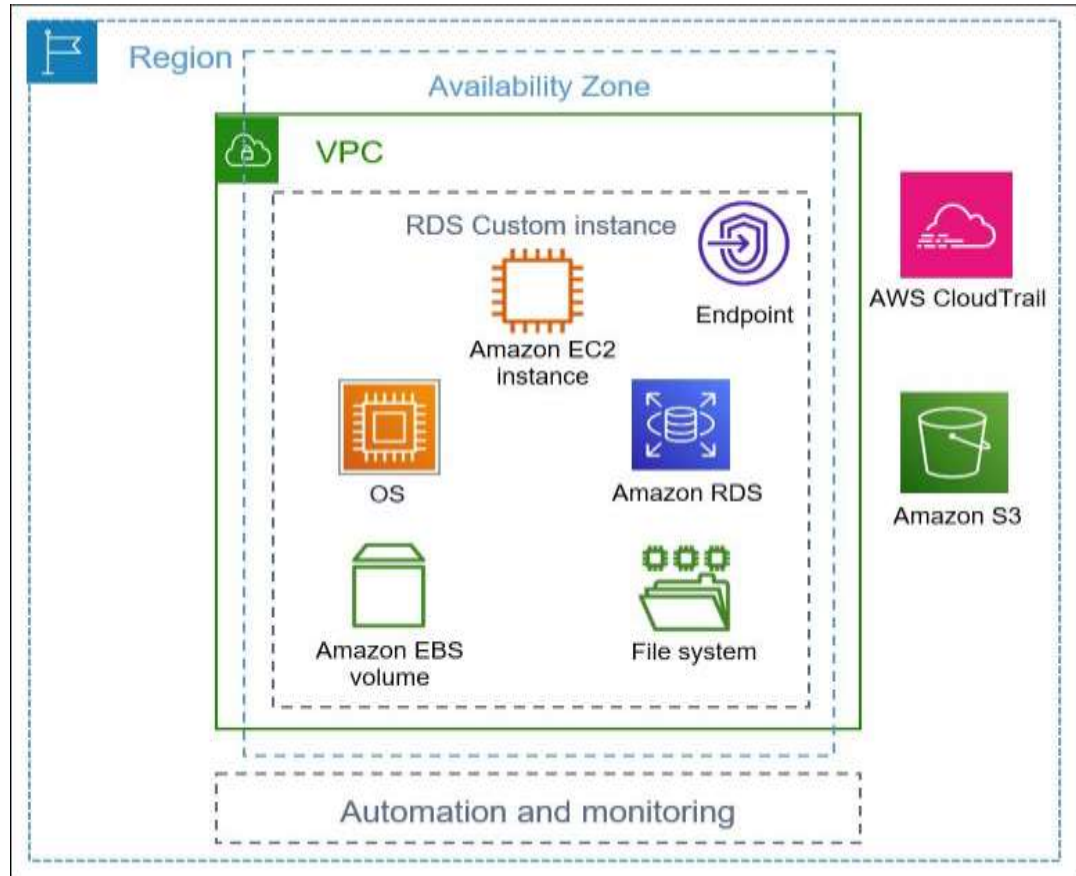
AWS RDS supports multiple database engines:

- MySQL
- PostgreSQL
- MariaDB
- Oracle
- SQL Server
- Amazon Aurora

Core Components of AWS RDS

- **DB Instance:** A virtual database server running a database engine.
- **Storage:** Persistent SSD-based storage attached to the instance.
- **Endpoint:** DNS address used to connect applications to the database.
- **Parameter Groups:** Configurations controlling database engine behavior.
- **Security Groups:** Firewall rules controlling inbound traffic.
- **Snapshots:** Point-in-time backups of database instances.
- **Subnet Groups:** Define VPC subnets where RDS instances can be deployed.

AWS RDS Architecture



Multi-AZ Deployment

Multi-AZ provides high availability by maintaining a standby replica in another Availability Zone.

Working

- Primary database handles traffic
- Standby replica remains synchronized
- Automatic failover during failure

Advantages

- High availability
- Reduced downtime
- Improved disaster recovery

Read Replicas

Read replicas improve application performance by distributing read workloads.

Uses

- Reporting systems
- Analytics queries
- Read-heavy applications

Key Points

- Asynchronous replication
- Multiple replicas supported
- Used mainly for scaling reads

Pricing and Costing

RDS pricing depends on:

- Instance type
- Storage
- Backup storage
- Multi-AZ deployment
- Data transfer

Creating the RDS Instance

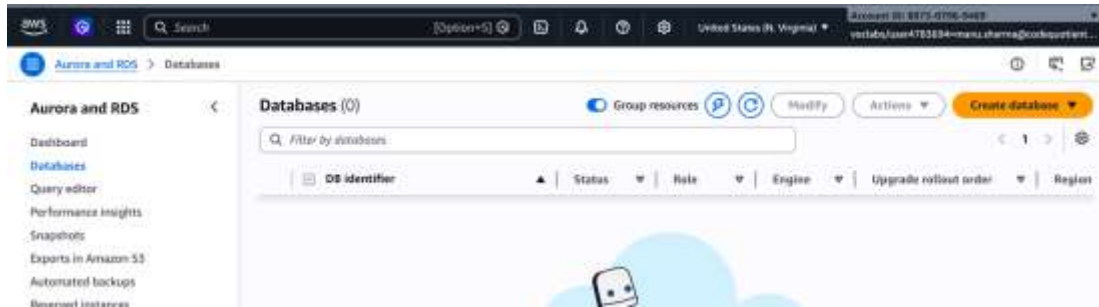
Content:

- Choose engine
- Template selection
- Credentials setup
- Instance class
- Storage configuration
- Connectivity settings
- Security group setup

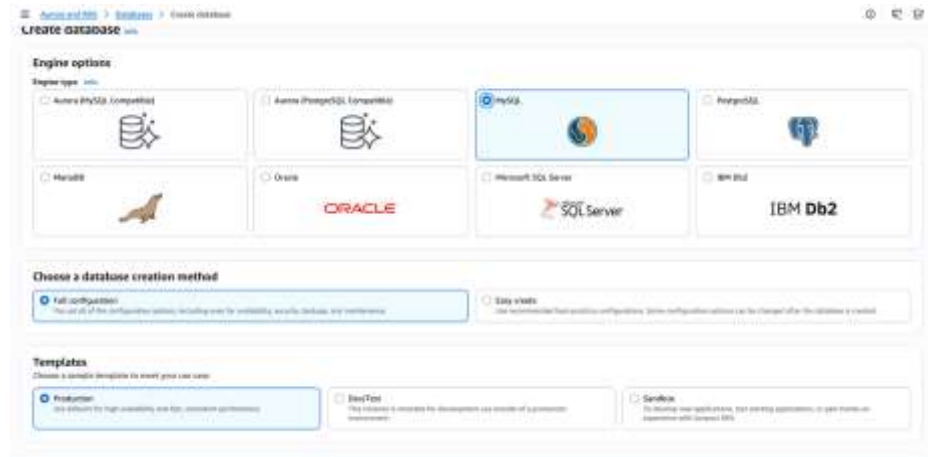
Important Settings

- Public access disabled
- Port 3306

1. Create a Database



2. Choose Engine and template according to your need(sandbox for learning)



3. Enable Public access (for testing purposes) and choose security group with inbound/outbound traffic on port 3306

The screenshot shows the 'DB subnet group' configuration page in the AWS RDS console. It includes sections for 'DB subnet group', 'Public access', and 'VPC security group (firewall)'. The 'Public access' section has the 'Yes' radio button selected. The 'VPC security group (firewall)' section has the 'Choose existing' button selected.

DB subnet group [Info](#)
Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

default-vpc-04e830f53c430538e
8 Subnets, 6 Availability Zones

Public access [Info](#)
☒ **Yes**
RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.
☐ **No**
RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

VPC security group (firewall) [Info](#)
Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

☒ **Choose existing**
Choose existing VPC security groups

☐ **Create new**
Create new VPC security group

4. Create your Database and get the endpoint link under Connectivity and security tab when the database's status is available

The screenshot shows the 'Connectivity & security' tab in the AWS RDS console. It displays the 'Database name' as 'mysql', the 'Endpoint type' as 'Instance endpoint', and the 'Endpoint' as 'database-1.cgp3y6yokg9g.us-east-1.rds.amazonaws.com'. The 'Port' is listed as '3306'.

Database name
mysql

Endpoint type
Instance endpoint

Additional configurations

Connectivity & security

Endpoint & port

Endpoint
database-1.cgp3y6yokg9g.us-east-1.rds.amazonaws.com

Port
3306

Connecting the Database via EC2

1. Launch an EC2 instance
2. Connect it via ssh or browser
3. In CLI

sudo yum update -y

sudo yum install mysql -y

mysql -h RDS-ENDPOINT -u admin -p

Connecting Database via MySQL Workbench

- Install MySQL Workbench
- Click on '+' and add a new connection
- Enter the username, Hostname and master password created during RDS database creation.
- Click on Test Connection.
- Database is connected Now.

The image shows two side-by-side screenshots. The left screenshot is from the Amazon RDS console, displaying the details for an instance named 'rds-mysql-10mintutorial'. The right screenshot is from the MySQL Workbench 'Connect to Database' dialog, showing the configuration for connecting to the RDS instance. Red boxes and arrows highlight the mapping of values between the two interfaces.

Amazon RDS Console Details:

- Summary:**
 - DB identifier: rds-mysql-10mintutorial
 - CPU: 7.80%
 - Role: Instance
 - Current activity: 0 Connections
- Connectivity & security:**
 - Endpoint & port:**
 - Endpoint: rds-mysql-10mintutorial.cx12hqsastfl.us-east-1.rds.amazonaws.com
 - Port: 3306
 - Networking:**
 - Availability Zone: us-east-1f
 - VPC: vpc-060e30a3d0f63f6
 - Subnet group: default-vpc-

MySQL Workbench 'Connect to Database' Dialog:

- Stored Connection:** (Empty)
- Connection Method:** Standard (TCP/IP)
- Parameters Tab:**
 - Hostname: rds-mysql-10mintutorial
 - Port: 3306
 - Username: masterUsername
 - Password: (Store in Keychain ... / Clear)
 - Default Schema: (Empty)

Red Boxes and Arrows:

- Red box around 'rds-mysql-10mintutorial' in the RDS console endpoint field points to the 'Hostname' field in MySQL Workbench.
- Red box around '3306' in the RDS console port field points to the 'Port' field in MySQL Workbench.
- Red box around 'masterUsername' in the RDS console username field points to the 'Username' field in MySQL Workbench.

Amazon RDS Console

MySQL Workbench

MySQL Workbench

TestDatabase x

File Edit View Query Database Server Tools Scripting Help

Administration Query 1 x

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup & Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Query 1 x

Limit to 1000 rows

```
1 • show databases;
```

2

Result Grid Filter Rows: Export: Wrap Cell Content:

#	Database
1	classicmodels
2	information_schema
3	mysql
4	performance_schema
5	sys

Object Info Result 1 x Read Only

Query Completed

Performing Database Operations

SQL Commands

```
CREATE DATABASE company;  
USE company;
```

```
CREATE TABLE employees(  
  id INT PRIMARY KEY,  
  name VARCHAR(50),  
  department VARCHAR(50)  
);
```

```
INSERT INTO employees VALUES  
(1, 'Rahul', 'IT');
```

```
SELECT * FROM employees;
```